

**MECE E3610 Materials and Processes in Manufacturing
Term Group Project Report**

Professor Sinisa Vukelic

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Race Car Manufacturing: Engineering Report

Group #10

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1 Introduction - Project Objective

In this project, our team designed and fabricated a model of a 1920's race car using a variety of manufacturing techniques explored throughout the course. Our team decided to produce a model of a 1924 Delage 2LCV [1].

The project included manufacturing techniques available to simulate an engineering environment through CAD 3D modeling, 3D printing, CAM, CNC Milling, silicone casting, and injection molding. Four main components were fabricated for this project: car body, cockpit, axles, and wheels. The molds manufactured and used for other components are shown in Figure 1.

After designing and creating the necessary components, the parts were assembled into a functional model to represent the reference design.

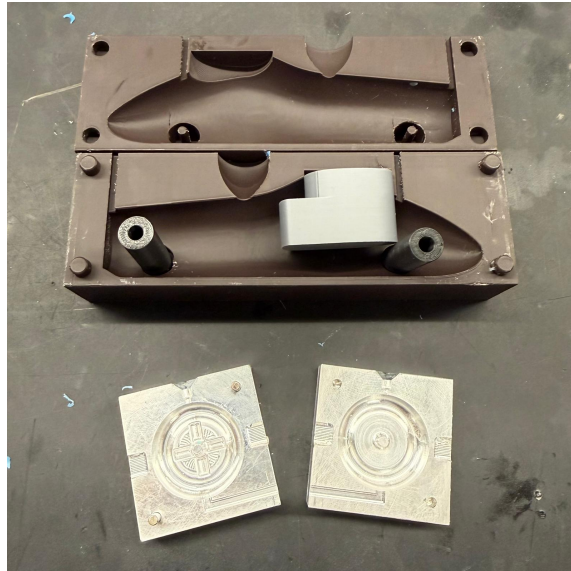


Figure 1: Molds used throughout the project for silicone casting and injection molding.

2 Manufacturing Process Overview

CAD Modeling and Design

The project started through a research and brainstorming stage to select a representative car from the 1920's period. The 1924 Delage 2LCV was selected. After exploring its dimensions and original design, a 3D model was made using SolidWorks to recreate the car's geometry. It was important for the car's body to have a "looping profile" where the height decreases from the center for casting purposes to avoid air pockets. First, the car body was modeled using the "lofting" feature. A separate part was designed for the cockpit. A cockpit spacer was made and, using the "combine" feature, a section of the car body was hollowed out. Finally, the axles and wheels were also designed as separate parts and their locations cut from the car body. All parts were put together as a SolidWorks assembly. The final CAD model is shown in Figure 2.

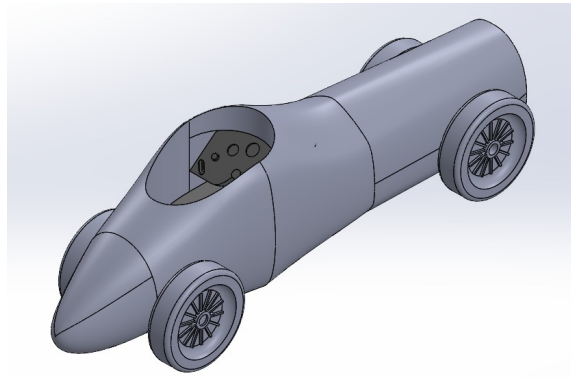


Figure 2: CAD Model of the 1924 Delage 2LCV race car.

Silicone Casting

The silicone casting process involved two separate steps: the design of the mold and the actual pouring and casting process. Silicone casting was used due to availability and safety.

Mold Design

Mold dimensions were set to be 6.25" x 2.25" x 2.25", so a rectangle with those dimensions was extruded to serve as the mold. Using the SolidWorks CAD file for

the car body (after applying the "combine" feature to subtract volume for cockpit and axle space), the car pattern was combined with the mold to subtract the volume that would get filled by the silicone to result in the car shape. A sprue, vents, and locator pins had to be designed to allow for casting as learned in class such that no air pockets, vacuums, or bubbles were formed.

Pouring and Solidification

The silicone casting process was mostly successful with minor changes that were made to perfect the design. Our original design was printed using standard print parameters which resulted in a semi-rough surface. Therefore, we reprinted vertically using fine settings to achieve a smooth finish. Additionally, since our cockpit had side walls, a locator triangle remained as part of the casting procedure which was removed and fixed by attaching the cockpit pattern to the "roof". We also had to shift our parting line which was originally off-centered. The sprue was also too large and in an inconvenient location to remove, so its location and dimensions were changed.

The cockpit pattern and axle holders were inserted into the mold where silicone would not reach to create a negative space. The two-part silicone was mixed together (65g for each part of the mixture). The mixture was placed in a vacuum to remove all air bubbles. Finally, it was slowly poured into the sprue until the cavity got filled. After at least six hours, the mold was pried open and the resulting body removed carefully. The sprue and vents were cut off as well as any imperfections or residue.

The casted car body used in the final product is shown in Figure 3.

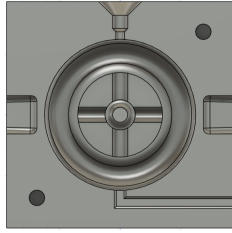


Figure 3: Final silicone cast for the car body.

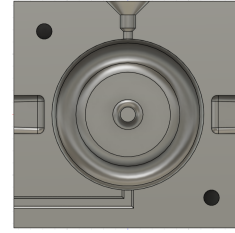
Injection Molding

Injection Mold Design

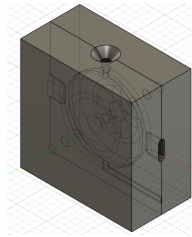
The injection mold was designed to be machined from a stock piece of aluminum using a CNC mill. The mold was created by subtracting features from the stock piece using the "combine" feature in Fusion 360. The wheel's geometry was captured using this technique, along with creating other features such as the injection nozzle hole and runner, pry cavities, alignment holes, and a small cooling vent. Next, the Computer-Aided Manufacturing (CAM) for the two mold halves was created within the Fusion 360 workflow. The following tools were used in the machining of the mold: 1/8 in ball end mill, 1/16 in ball end mill, center drill, and #30 drill. The CAM parameters were assigned based on the provided specifications so that the piece could be machined safely and accurately. A simulation of the CAM toolpaths was performed to ensure the machining was safe to perform.



(a) Front Mold wheels with spokes.



(b) Back Mold



(c) Isometric Mold

Figure 4: Fusion360 Models of the Injection Molding Mold.

CNC Machining

After completing the CAM for the wheel mold, the G-Code was generated and exported. The G-Code was imported into the Tormach770M milling machine. Two solid halves of aluminum were set and clamped into the machine. The top-left corner was found and used as a reference. The code was run and tools were changed as

needed throughout the process. The mechanical engineering teaching lab procedure was followed [2]. The resulting mold is shown in Figure 5.



Figure 5: Mold used for injection molding after machining.

Injection Molding Procedure

Using the injection molding machines in the Mechanical Engineering Teaching Lab, along with the previously manufactured mold, the wheels were created. Plastic pellets were inserted into the machine, melted, and injected into the clamped mold's sprue. Multiple attempts were needed to properly align the mold to the nozzle, but after time and practice, the process became more straightforward. It is mainly tedious due to the time needed between attempts, but all wheels were completed. Since the shop ran out of PLA, Polypropylene had to be used, so a total of six wheels were completed to ensure a complete set of same-material wheels. The wheels were also sanded for a smoother finish. Figure 6 shows the final result of the wheels after injection molding.

Assembly and Completion

We cut the axles to desired length and added a chamfer to each end using the machine shop's disc sander. For assembly, one wheel was press-fit onto each axle by placing the wheel face-down on a table and hammering the axle through from above. The custom 3D-printed axle holders, perfectly contoured to the shape of our car body, were then inserted into the car body. These reduced the clearance and kept the axles from excessive wiggling. The axles were then fed through the front and rear holders,



Figure 6: Resulting wheels from injection molding.

and the second wheel on each axle was hammered on with a cloth placed over the wheel to avoid damaging it during the process. Finally, the cockpit was slid into its position in the car body.

The final product is shown in figure 7.



Figure 7: Final Assembled Race Car Model.

3 Conclusions - Project Reflection and Analysis

What Went Well

Overall, the project went very smoothly. Our team worked great together to collaborate and produce a product that we were all satisfied with. We delegated roles, and every member contributed meaningfully to the project. The design phase was smooth, as we were able to follow the materials provided to create an accurate model of the car. We were able to quickly learn the fabrication methods and produced multiple consistent results with the silicone casting and injection molding.

Challenges Faced

Our team faced some difficulties fleshing out the final design of the car body mold. We produced a total of three different molds and cast silicone in each. With every iteration, we improved details like the sprue size and location, parting line, and post-processing until we were left with a final clean cast that was free of imperfections. We also faced challenges in customizing our wheels to include extra details, such as spokes. It was difficult to include these additional features in the CAM to fit within the workflow provided in the tutorial.

An example of this iteration process was the silicone mold which, as stated, was made three times. Figure 8 shows the three different molds with the first iteration at the bottom and the latest iteration at the top.

Lessons Learned

This group project highlights the importance of being detail-oriented when working in a team. We learned quickly that communication was vital to working effectively together, and being detailed with design choices was crucial for a great end product. We also learned how testing and iterating on designs is an essential part of the manufacturing process.

Improvements for Future Work

If we were to continue this project in the future, the design could be improved to include more detailed features and to make parts more compatible with each other. For example, our cockpit sits slightly too low within the car body, allowing the silicone to wrap around the top and slightly block some of the included details.



Figure 8: Three mold iterations used. Bottom being the first mold and the latest being the topmost mold.

Additionally, more attention to tolerances, such as those for the axle holes and wheel attachments, would improve the end product and reduce post-processing work.

Final Takeaways

This project was a great opportunity to gain hands-on experience with a variety of manufacturing techniques. We strengthened our skills through the design process by working with CAD and CAM, and better understood the decisions that are made when considering the manufacturability of different parts. We gained a better appreciation for the factors that influence design decisions, such as materials, machining/tooling, and casting/mold-making. It was rewarding to see each of our group members contribute and collaborate, as our communication and teamwork skills were reinforced through working together on this project.

A Final Product Images

The finished product can be seen in both Figure 7 and more views are seen as part of the appendix in Figure 9



Figure 9: Different views of the finished car model.

References

- [1] Exclusive Car Registry. Delage 2LCV details. <https://exclusivecarregistry.com/details/delage/2lcv/81755>, 2026. Accessed: March 26, 2026.
- [2] Columbia University Mechanical Engineering. The Mechanical Engineering Teaching Laboratory. <https://teachinglab.me/>. Accessed: April 20, 2026.